

Abhishek Tiwari

Nagpur, India | +91-9595266571 | abhishektiwari00172@gmail.com | [LinkedIn](#) - www.linkedin.com/in/aaabhihektiwari |

[GitHub](#) - <https://github.com/tiwaria00172>

OBJECTIVE

Second-year B.Tech Computer Science student (AIML) (CGPA: 9.3/10) with a strong foundation in Machine Learning, AI systems, and full-stack development. Deeply interested in Brain-Computer Interfaces and decoding neural signals to infer human intent. Currently building systems at the intersection of machine learning, real-time data, and human-centred applications.

EDUCATION

B.Tech – Computer Science & Engineering (Artificial Intelligence and Machine Learning) 2024 – 2028 (Expected)
Ramdeobaba University (RBU), Nagpur | CGPA: 9.3 / 10.0

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, HTML/CSS

ML & Data: Scikit-learn, NumPy, Pandas, Matplotlib, pattern recognition, time-series & behavioural analysis

Hardware: Arduino, ESP32, HC-SR04, I2C OLED, DHT sensor, L298N motor driver

Tools: FastAPI, Firebase, Git, Jupyter Notebook, Google Colab, VS Code

PROJECTS

EEG Headband Prototype – Entrepreneurship Project 2026

- Conceptually designed a dry-electrode EEG headband as part of an entrepreneurship course, covering electrode placement strategy, signal acquisition circuitry architecture, and wearable form factor; theoretically addressed noise rejection, muscle artefact, and skin-electrode impedance challenges.
- Presented product pitch, market positioning, and technical feasibility report to a panel, evaluating potential manufacturability and pathways toward a consumer neurofeedback device.

FitSync – AI Health & Diet Companion App (IN PROGRESS) 2026 – Present

- Building a health app that ingests multi-modal signals (nutrition, activity, sleep, biometrics) and uses ML to generate personalised recommendations — methodology directly applicable to multi-channel biosignal fusion.
- Integrating anomaly detection on user health baselines; full-stack FastAPI + Firebase architecture built to support future wearable sensor integration.

HabitFlow – AI Behaviour Analytics App 2025 – Present

- Developing a student productivity application featuring AI-driven habit analytics and recommendations, including a machine learning pipeline for behavior-based schedule optimization, built as a full-stack system using FastAPI, Firebase, and interactive dashboards to enhance user engagement and retention.

Obstacle Detection & Avoidance Bot – Arduino 2025

- Real-time autonomous system with multi-sensor fusion and embedded C++ control loops; gained practical understanding of sensor noise, calibration drift, and signal variability

QuizCraft – NLP-Based MCQ Generator 2025

- NLP keyword extraction from unstructured text into structured outputs; cognitive load-aware UX design reflecting awareness of neural engagement mechanisms.

RELEVANT COURSEWORK

Artificial Intelligence • Machine Learning Foundations • Embedded Systems (ESP32, Arduino) • Data Structures & Algorithms • Computer Networks • Database Management Systems

HACKATHONS & LEADERSHIP

Smart India Hackathon – RBU Internal Round 2025 | Shortlisted 1

- Co-led a 4-person team; oversaw technical architecture, feasibility scoping, and full demo delivery to panel.

Inter-College Hackathon – SB Jain Institute 2025 | Shortlisted Round 1

- Led fintech team building an offline P2P payment tool; directed technical development and cross-functional coordination.

EXTRACURRICULARS

3D Artist – College Technical Club | 3D models, animations & event posters using Blender.

Member – Kaagaz Open Mic Community | Event management & community building, Nagpur.